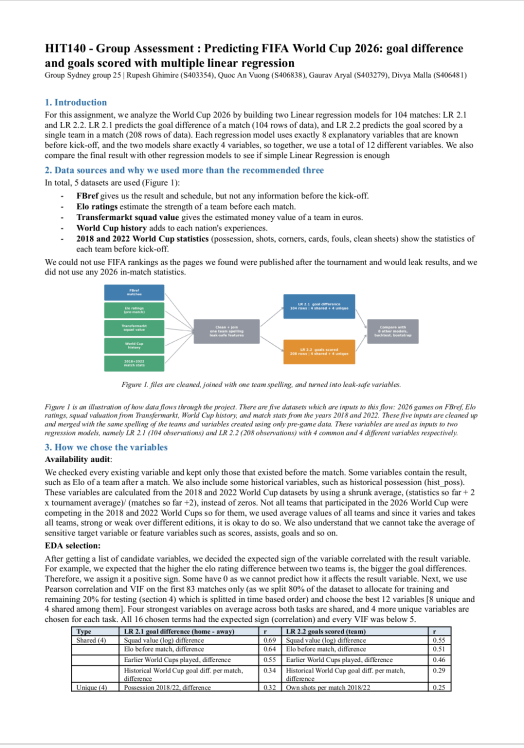
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HIT140 - Group Assessment : Predicting FIFA World Cup 2026: goal difference and goals scored with multiple linear regression

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1. Introduction

For this assignment, we analyze the World Cup 2026 by building two Linear regression models for 104 matches: LR 2.1 and LR 2.2. LR 2.1 predicts the goal difference of a match (104 rows of data), and LR 2.2 predicts the goal scored by a single team in a match (208 rows of data). Each regression model uses exactly 8 explanatory variables that are known before kick-off, and the two models share exactly 4 variables, so together, we use a total of 12 different variables. We also compare the final result with other regression models to see if simple Linear Regression is enough

2. Data sources and why we used more than the recommended three

In total, 5 datasets are used (Figure 1):

- **FBref** gives us the result and schedule, but not any information before the kick-off.
- **Elo ratings** estimate the strength of a team before each match.
- **Transfermarkt squad value** gives the estimated money value of a team in euros.
- **World Cup history** adds to each nation's experiences.
- **2018 and 2022 World Cup statistics** (possession, shots, corners, cards, fouls, clean sheets) show the statistics of each team before kick-off.

We could not use FIFA rankings as the pages we found were published after the tournament and would leak results, and we did not use any 2026 in-match statistics.

Figure 1. files are cleaned, joined with one team spelling, and turned into leak-safe variables.

Figure 1 is an illustration of how data flows through the project. There are five datasets which are inputs to this flow: 2026 games on FBref, Elo ratings, squad valuation from Transfermarkt, World Cup history, and match stats from the years 2018 and 2022. These five inputs are cleaned up and merged with the same spelling of the teams and variables created using only pre-game data. These variables are used as inputs to two regression models, namely LR 2.1 (104 observations) and LR 2.2 (208 observations) with 4 common and 4 different variables respectively.

3. How we chose the variables

Availability audit

We checked every existing variable and kept only those that existed before the match. Some variables contain the result, such as Elo of a team after a match. We also include some historical variables, such as historical possession (hist_poss). These variables are calculated from the 2018 and 2022 World Cup datasets by using a shrunk average, (statistics so far + 2 x tournament average)/ (matches so far +2), instead of zeros. Not all teams that participated in the 2026 World Cup were competing in the 2018 and 2022 World Cups so for them, we used average values of all teams and since it varies and takes all teams, strong or weak over different editions, it is okay to do so. We also understand that we cannot take the average of sensitive target variable or feature variables such as scores, assists, goals and so on.

EDA selection:

After getting a list of candidate variables, we decided the expected sign of the variable correlated with the result variable. For example, we expected that the higher the elo rating difference between two teams is, the bigger the goal differences. Therefore, we assign it a positive sign. Some have 0 as we cannot predict how it affects the result variable. Next, we use Pearson correlation and VIF on the first 83 matches only (as we split 80% of the dataset to allocate for training and remaining 20% for testing (section 4) which is splitted in time based order) and choose the best 12 variables [8 unique and 4 shared among them]. Four strongest variables on average across both tasks are shared, and 4 more unique variables are chosen for each task. All 16 chosen terms had the expected sign (correlation) and every VIF was below 5.

Type	LR 2.1 goal difference (home - away)	r	LR 2.2 goals scored (team)	r
Shared (4)	Squad value (log) difference	0.69	Squad value (log) difference	0.55
	Elo before match, difference	0.64	Elo before match, difference	0.51
	Earlier World Cups played, difference	0.55	Earlier World Cups played, difference	0.46
	Historical World Cup goal diff. per match, difference	0.34	Historical World Cup goal diff. per match, difference	0.29
Unique (4)	Possession 2018/22, difference	0.32	Own shots per match 2018/22	0.25

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